

Paper 1B drill: which error, and which way?

Sixteen items. Classify it, then commit to a direction.

IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **26 marks**

A *random* error scatters repeated readings either side of the true value, so averaging helps. A *systematic* error shifts every reading the same way, so averaging does not help at all and the result stays wrong however many times you repeat it.

A *systematic or random?*

State which, in one word each.

- A1. A top-pan balance reads 0.4 g with nothing on it. [1]
- A2. A stopwatch is started slightly late or slightly early each time, unpredictably. [1]
- A3. A metre rule has its first centimetre worn away and the student measures from the broken end. [1]
- A4. The mains voltage fluctuates during an experiment. [1]
- A5. A student reads a liquid level from slightly above the meniscus every single time. [1]
- A6. The diameter of a wire is measured in one place only, but the wire is not perfectly uniform. [1]

B *now commit to a direction*

For each, state whether the calculated result comes out *too high*, *too low*, or *scattered but not shifted*, and explain in one sentence.

- B1. A micrometer reads +0.03 mm when closed. It is used to measure a wire's diameter, and the wire's cross-sectional area is then calculated. [2]
- B2. A student measures g by timing a ball falling a measured height, but consistently starts the stopwatch a moment *after* releasing the ball. [2]
- B3. In the same experiment, air resistance is ignored. [2]
- B4. A thermometer bulb is not fully immersed in the hot liquid it is measuring. [2]
- B5. An immersion heater warms a metal block, and some thermal energy escapes to the room. The specific heat capacity is then found from $c = VIt/(m \Delta T)$. [2]
- B6. The length of a pendulum is measured to the *top* of the bob rather than to its centre, and g is found from $T = 2\pi\sqrt{L/g}$. [3]

C *what to do about it*

- C1. Explain why repeating a measurement many times and averaging does nothing at all for a zero error. [2]
- C2. A student times one oscillation of a pendulum twenty times and averages. A classmate times twenty oscillations once and divides by twenty. State which is better and why. [3]
- C3. One reading in a set of eight lies far from the trend. State what you should do, and what you must not do. [2]

Answers

A *systematic or random?*

- A1. Systematic. A zero error offsets every reading identically.
 - A2. Random. The sign of the error varies unpredictably.
 - A3. Systematic. Every length is wrong by the same missing centimetre.
 - A4. Random. The fluctuation is as often up as down.
 - A5. Systematic. Always viewing from the same wrong angle shifts every reading the same way. (Parallax is only random if the viewing angle varies.)
 - A6. Random, in effect: measuring at a different place each time scatters the readings, which is why the fix is to measure at several points and average.
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B *direction*

- B1. **Too high.** Every diameter is 0.03 mm larger than the truth, and since $A = \pi d^2/4$ the area is too big, roughly twice as far out in percentage terms because the diameter is squared.
 - B2. **Too high.** Starting late records a time shorter than the real fall, and $g = 2h/t^2$ has t in the denominator, so a smaller t gives a larger g .
 - B3. **Too low.** Air resistance opposes the fall, so the ball takes longer than it would in free fall, and a larger t in $g = 2h/t^2$ gives a smaller g .
 - B4. **Too low.** The exposed stem sits in cooler air, so the thermometer reads below the true liquid temperature.
 - B5. **Too high.** The block receives less energy than the VIt assumed, so it rises by less than it should; since ΔT is in the denominator of $c = VIt/(m \Delta T)$, a smaller measured rise gives a larger c . (Students almost always guess too low here, on the reasoning that losing energy must lose you something. What is lost is the temperature rise, and it sits underneath.)
 - B6. **Too low.** Measuring to the top of the bob gives a length shorter than the true pendulum length to the centre of mass. From $T = 2\pi\sqrt{L/g}$, rearranged $g = 4\pi^2 L/T^2$, an underestimate of L gives an underestimate of g . Note this one cannot be fixed by repeating: every single measurement carries the same error, which is what makes it systematic.
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C *what to do about it*

- C1. Averaging removes error that is as often positive as negative. A zero error is the same size and the same sign every time, so it survives the average untouched: the mean of a hundred readings each 0.03 mm too big is still 0.03 mm too big. It has to be removed by subtraction, or by zeroing the instrument first.

- C2. Timing twenty oscillations once is better. The reaction-time uncertainty of roughly 0.2 s is incurred *once* for the whole run and is then divided by twenty along with the time, giving about 0.01 s per period. Timing single oscillations incurs that same 0.2 s on every one of the twenty measurements, and averaging twenty readings only improves it by a factor of about $\sqrt{20} \approx 4.5$, not 20. Same effort, roughly four times worse.
- C3. Investigate it and say so. Check the reading was not simply written down wrongly, repeat that measurement if you still can, and if you exclude it from the line of best fit, state clearly in your working that you have done so and why. What you must not do is delete it silently, or keep it and let it drag the gradient while pretending the fit is good. An anomaly is evidence about the experiment, not an embarrassment to hide.